

Escalate or Answer: Confidence-Triggered Human Escalation in Agentic RAG Improves Expert Trust Calibration and Novice Learning

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Abstract

Retrieval-augmented generation over proprietary corpora answers fluently even when retrieval is wrong, and fluency inflates trust independently of accuracy. We build one system manipulation and run two experiments on it. System A escalates to a human at seeded points of low-confidence or contradictory retrieval, with an escalation policy modeled on minimal proactivity design patterns; System B never escalates and always answers from its top retrieval. Experiment 1 asks whether domain experts grading fixed outputs calibrate trust better and choose more defensibly under System A. Experiment 2 asks whether novices onboarding to an invented internal tool score higher on an unassisted next-step post-test under System A. The result tells designers whether visible uncertainty is worth its interruption cost in agentic RAG.

Keywords

agentic AI; retrieval-augmented generation; trust calibration; human-in-the-loop; learning assistance

1 Introduction and Motivation

Retrieval-augmented generation (RAG) is a popular way to put a large language model (LLM) on top of an organization’s private documents. An issue that commonly arises in this setting is the output being a confident, well-formed answer built from the wrong context. Proprietary corpora is exactly where this is most likely and least detectable (documents contradict each other, deprecated information is left in place, the reader has no external source to check against).

Two different groups handle that failure differently. Domain experts must decide how much to trust each output, and agentic reasoning’s fluency can raise that trust without any change in underlying accuracy [5]. Novices onboarding with such a system face a second problem: richer training materials inflated participants’ confidence that they would remember correctly with no change in their actual ability to identify the next step [2].

An obvious solution is to let the system stop and ask a human to intervene when its own confidence is low. Agentic RAG architectures already describe this as a design pattern [4], but whether human intervention actually improves expert trust calibration or novice learning is an open empirical question where one must consider the cost of escalation, how much it interrupts the flow, and the fact that it may simply be ignored.

2 Related Work

Agentic RAG and human oversight. Mishra et al. [4] survey agentic RAG architectures and describe a human-in-the-loop pattern in which human oversight is a callable action. “This pattern models human oversight as a callable API within the action space. When epistemic uncertainty exceeds a defined threshold, the policy pauses execution to request disambiguation or supervision.” They use Expected Calibration Error among the evaluation criteria for this system and leave adaptive escalation as an open direction. Their non-escalating configuration is the baseline this proposal adopts for System B.

When and how a system should interrupt. Sánchez-Adame and Mendoza [7] discuss Minimal Proactivity Policies (MiPP), a contract for system-initiated interruptions where initiations should be bounded and reversible, expose an explicit provenance cue naming the source that justified the interruption, and must default to silence rather than speculation when required inputs are missing or ambiguous. MiPP inspires the policy shape for System A’s escalation branch.

Trust calibration and its measurement. Möhle et al. [5] define trust calibration as the alignment between a user’s trust and the system’s actual capability on a task. They find that stated accuracy shifts trust more strongly when the accompanying reasoning is convincing. This is therefore not a uniform trust booster but a conditioning cue, which is why this proposal measures graded accuracy and revealed choice rather than self-reported confidence. Chakraborti et al. [1] supply the procedure: participants took the role of an external commander, evaluated plans produced by an internal robot, and rated them optimal or suboptimal, with end-of-study trust items on whether the robot could be left to work on its own.

Measuring the impression of learning. Clegg et al. [2] found that multimedia and immersive training materials raised participants’ confidence when asked if they would remember how to assemble objects even though their measured ability to identify the critical next step did not improve. This dissociation is the effect a fluent, never-escalating assistant is predicted to produce, and it motivates a task-embedded post-test instead of a declarative quiz.

Uncontaminated evaluation corpora. Kirchenbauer et al. [3] generate fictional documents and validate them by prompting the generator model blind (no access to fictional context) and then informed (access to fictional context), discarding items answerable without the document. This makes it possible to study retrieval over

a "proprietary" corpus without the model's pretraining knowledge confounding the result.

The gap. The escalation pattern is documented [4] and a policy for it is specified [7], but neither has been tested as an independent variable against a calibration outcome or a learning outcome on a controlled proprietary corpus. That intersection is where this project sits.

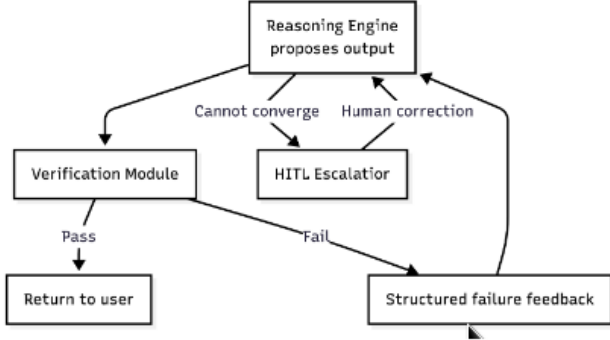


Figure 1: The Human-in-the-Loop escalation loop, taken directly from Mishra et al. (2026, their Fig. 5). System A’s policy for this escalation branch is modeled on MiPP (Sánchez-Adame and Mendoza, 2026); System B disables the branch and always proceeds to Final Response.

3 Approach

We build one manipulation and reuse it across two experiments on the same corpus.

System A (escalating). At seeded points of low-confidence or contradictory retrieval, System A pauses and requests human confirmation before proceeding. Its escalation policy is modeled on MiPP: the pause is brief and immediately reversible, it carries an explicit provenance cue naming the source that triggered it, and it defaults to silence rather than speculation when evidence is insufficient [7].

System B (non-escalating). System B always answers from its top retrieval, with no pause and no provenance cue, matching the anchor paper’s non-escalating baseline [4].

These three papers are design guides, not specifications this project is bound to reproduce in full; a three-month build-and-run window means each is adapted to what two people can implement and run.

The manipulation satisfies the field’s chatbot test: a study whose result would be unchanged with the agency removed has no independent variable [6]. Here, removing the escalation branch is the control condition, so agency is the variable rather than decoration. This is an Agentic-AI human-centered computing project: the contribution is an empirical result about how an autonomous decision to defer to a human changes human trust and human learning, not a language model artifact.

4 System Vision and Technology

Both experiments share one interface shell so the escalation policy is the only thing that varies. A task pane on the left holds the current item (an output to grade in Experiment 1, a checklist step in Experiment 2). The center pane holds the assistant’s retrieval-and-response output with its cited sources. A decision pane on the right collects the participant’s response and advances the session. Under System A, a seeded low-confidence or contradictory retrieval replaces the answer with an escalation prompt that names the conflicting source and offers a bounded, reversible choice; under System B the same item renders as a direct answer with no pause and no provenance cue.

Participants work at a desk on a desktop browser, with printed reference documents available in the expert condition. Instruments (forced choice, NASA-TLX, free-text justification) appear as full-screen steps between tasks rather than inline, so they do not compete with the manipulation. The retrieval stack is Python; responses are generated in advance and replayed fixed, so both conditions present identical content except at the seeded escalation points.

Figure 2: Mock-up for System A’s digital escalation interface upon mis-retrieval of information for a given question.

5 Evaluation Plan

H1 (Trust). Among domain experts grading individual retrieval-and-response outputs from System A and System B on a shared proprietary corpus, (a) experts’ grades will track true output accuracy more closely for System A than for System B, and (b) when asked at the end of the session which system they would choose for an upcoming task and why, experts will choose System A more often and justify it by citing verifiable evidence rather than confidence or fluency alone.

H2 (Learning). Among novice engineers completing a short onboarding checklist for the same proprietary tool with System A or System B, System A will produce a higher score on an unassisted, task-embedded next-step post-test.

Variables. The independent variable in both experiments is escalation policy (System A escalates vs. System B never escalates),

manipulated within subjects with counterbalanced order. The dependent variables are, for Experiment 1, the correlation between an expert’s grade and the output’s true accuracy (calibration), plus final forced choice and the thematic code of its justification; and for Experiment 2, accuracy on the unassisted next-step post-test, plus completion time. Table 1 gives the full design (see last page).

Analysis. Paired comparison per experiment (Wilcoxon signed-rank or paired t -test after a normality check), with effect size and confidence interval; thematic analysis of Experiment 1’s free-text justifications using the course’s coding method [6].

Human participants. Both experiments involve human participants. We will file a CSU IRB protocol covering both, since we intend the results to be publishable.

6 Deliverables

- A generated proprietary corpus for one invented internal tool (README, changelog, wiki page, deprecated guide) with span-constrained QA pairs and a passed blind-versus-informed contamination check.
- A Python retrieval stack with two configurations, System A (confidence-triggered escalation) and System B (no escalation), and a fixed response set shared by both experiments.
- Experiment 1’s static grading interface and Experiment 2’s scripted two-arm prototype, both with instrumented logging.
- An approved CSU IRB protocol covering both experiments.
- Logged trial data from $N \approx 15$ –20 experts (Experiment 1) and $N \approx 15$ –20 novices (Experiment 2), plus coded free-text justifications.
- A paper draft with Method and Results for both experiments, and a demo video of the two conditions side by side.

7 Conclusion

We are testing whether an agentic RAG assistant that stops and asks a human when its own retrieval is weak produces more trusting experts and better-learning novices versus an agentic RAG assistant that always answers regardless of the quality of the context retrieved. The intervention is a single escalation policy, held constant across two experiments on one controlled proprietary corpus, so both results are aligned with the same design decision. The expectation is a working two-condition prototype, an uncontaminated corpus, and paired quantitative results.

Use of AI

Generative AI was used to assist with literature search, to critique the argument for weaknesses, and for grammar and Latex formatting fixes.

References

- [1] Tathagata Chakraborti, Sarath Sreedharan, Sachin Grover, and Subbarao Kambhampati. 2018. Plan Explanations as Model Reconciliation: An Empirical Study. *arXiv preprint arXiv:1802.01013* (2018). <https://arxiv.org/abs/1802.01013>
- [2] Benjamin A. Clegg, Andrew Karduna, Emily Holen, Jonathan Garcia, Matthew G. Rhodes, and Francisco R. Ortega. 2022. Multimedia and Immersive Training Materials Influence Impressions of Learning but Not Learning Outcomes. In *Proceedings of the Interservice/Industry Training, Simulation and Education Conference (IITSEC)*. Orlando, FL, USA.

- [3] John Kirchenbauer, Nikhil Mongkolkeha, Yuxin Wen, Tom Goldstein, and Daphne Ippolito. 2026. FictionalQA: A Dataset for Studying Memorization and Knowledge Acquisition. *arXiv preprint arXiv:2506.05639* (2026). <https://arxiv.org/abs/2506.05639>
- [4] Subash Mishra, Sandesh Niroula, Umesh Yadav, Dipesh Thakur, Sujan Gyawali, and Sagar Gaire. 2026. SoK: Agentic Retrieval-Augmented Generation (RAG): Taxonomy, Architectures, Evaluation, and Research Directions. *arXiv preprint arXiv:2603.07379* (2026). <https://arxiv.org/abs/2603.07379>
- [5] Marc H. Möhle, Maximilian Zall, Tobias Thorwart-Gumpert, Niklas Elbert, Frédéric Thiesse, and Christoph M. Flath. 2026. “Trust Me, Bro!” On the Influence of Agentic Support on Trust Calibration in AI-Assisted Decision Making. In *Proceedings of the 28th International Conference on Human-Computer Interaction (HCII 2026)*. Springer, Cham, Switzerland, 472–488.
- [6] Francisco R. Ortega. 2026. Research Problems and Experiments. CS 567 course slides, Colorado State University.
- [7] Luis Martín Sánchez-Adame and Sonia Mendoza. 2026. MiPP: Minimal Proactivity Policies for Timely, Low-Intrusion Proactivity. In *Proceedings of the 28th International Conference on Human-Computer Interaction (HCII 2026)*. Springer, Cham, Switzerland, 363–382.

	Experiment 1: Trust (H1)	Experiment 2: Learning (H2)
Population	Domain experts grading system output	Novice engineers (new hires) onboarding
Task	Grade a fixed set of retrieval-and-response outputs, half from System A and half from System B	Complete a short onboarding checklist for the invented tool, using System A or System B
Procedure	Blind, randomized-order grading of outputs fixed in advance [1, 5]; end-of-session forced choice of A or B for an upcoming task, with a free-text reason	Pull-based query loop (Section 4); two or three checklist items seeded with low-confidence or contradictory retrieval
Primary measure	Grade-accuracy correlation per system (calibration)	Unassisted, task-embedded next-step post-test accuracy
Secondary measures	Final choice (A/B); thematic code of the justification (evidence-based vs. confidence/fluency-based); competence/benevolence/integrity trust scale [5]	Completion time; NASA-TLX (optional)
Design	Within-subjects, counterbalanced order, $N \approx 15-20$	Within-subjects, two-level independent variable, $N \approx 15-20$
Session length	~15–20 min; no live model prototype needed, outputs are pre-recorded	~30–45 min

Table 1: The two experiments, sharing System A (escalates, policy modeled on MiPP [7]) versus System B (never escalates [4]) on the same proprietary corpus.